

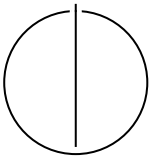
SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
FACULTY OF INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Decoding Music Perception: EEG-Based Genre Classification

Lukas Alex Jochim





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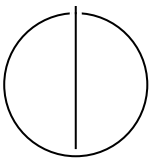
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Decoding Music Perception: EEG-Based Genre Classification

Dekodierung der Musikwahrnehmung: EEG-basierte Genre-Klassifizierung

Author:	Lukas Alex Jochim
Examiner:	Prof. Dr. Björn Schuller
Supervisor:	MSc. Simon Pistrosch
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I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, 01.09.2025 Lukas Alex Jochim

Acknowledgments

Transparency in the use of AI tools

While writing this thesis, Grammarly was utilized to ensure high-quality grammar, style, clarity, and coherence in the content of each chapter. Furthermore, DeepL was used for translation purposes. During the development phase, ChatGPT and Claude Sonnet were used to generate simple code snippets and to gather first ideas for implementing certain functionalities. However, everything generated by one of those tools has been carefully evaluated by me to ensure correctness.

Abstract

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1 Introduction

1.1 Motivation

Maybe also mention here the gap between MGR and neurosciences more well, since it will be brought up in the Research goals section The human brain is one of the most complex and important systems within the human body. The brain manages various functionalities like emotion, memory, attention and sensory processing, to name a few. Apart from that, the brain plays a significant role in how we process musical pieces. Independent of culture and age, music is a universal tool to influence one's emotions, concentration and cognition. Furthermore, the brain gives us the unique ability to retrieve many different features out of music, like melody extraction or the identification of an artist or a specific genre. Depending on the style of music, the brain reacts differently. However, although the brain is central to cognitive processes such as listening to music, the core mechanisms remain unknown and are currently black boxes.

To enlighten these black boxes, medical advancements like functional Magnetic resonsance imaging (fMRI) or Electroencephalography (EEG) arrived, which can measure brain activity. Especially EEG has aroused high interest since they directly measure the neural activity and can be done non-invasive with a little overhead in setup. This is why combining EEG and music has the ability to help better understand how our brain functions. By further adding means of Machine Learning (ML), a whole new interdisciplinary field opens up, bridging the gap between neurosciences and Computer science.

There is a growing research interest in the affective influence of music on the brain. However, classifying different music genres based on EEG signals remains underrepresented. This is exactly where this thesis aims to close the gap, by investigating whether it is possible to classify various musical genres depending on the EEG signals acquired using state-of-the-art ML techniques.

The research results have high potential in the domain of medical computer science. Especially with the growing investigations on Brain Machine Interfaces (BMIs), adaptive music therapy, personalised music recommendations or neurofeedback systems may become possible. Additionally, entertainment technologies can become more user-centred, facilitating unique experiences that impact many sectors like education and

mental health.

1.2 Research goals

1.3 Outline

Add that you compare classification results with the ones from the DEAP dataset

To answer the research questions mentioned above, it is vital to have a straightforward research approach, depicted by the structure of this thesis. In the following, a short thesis outline is given, focusing on structure and content in the succeeding chapters.

At first, a theoretical background on EEG signals is given in Chapter 2, building the basis of understanding for all further chapters. After covering the biological procedures behind the generation of EEG signals in the brain, it is described how one can infer the state of the brain based on EEG. In that case, the most prominent brain rhythms are emphasized, describing their impact on the brain and their influence on the measured signal. After the medical background behind the properties of EEG signals, the focus moves towards the three key steps when planning an experiment involving EEG with the classification background. Firstly, the theory behind measuring the EEG data is explained. This section explains what components an EEG system consists of, alongside standard practices regarding the measurement. With the acquisition in mind, the following section deals with preprocessing steps in EEG data, especially on how to deal with artifacts and noise in the raw signal. Once the different types of artifacts appearing have been clarified, different approaches regarding artifact handling are outlined and explained. Last but not least, the last section of the theoretical background covers feature extraction methods in the preprocessed data. Mathematical concepts behind certain extraction functions across all three relevant domains (time-domain, frequency-domain, time-frequency-domain) are broadly described alongside explanations on why these features should be considered.

Chapter 3 then examines the current state of the art in research regarding Music Genre Recognition (MGR) with a focus on EEG. Before investigating research on EEG, a short overview of how one can perform MGR on audio features is provided. This is important to cover as the brain conducts a form of MGR based on the audio signal. Then, a general review is given on EEG datasets and studies that use music to generate the signal. When showcasing these studies, a special focus is given on the experiment setup/environment, measurement setups, and preprocessing techniques. Furthermore, current research is presented on MGR with EEG signals. However, since this is still a relatively young topic in research, this section is accompanied by studies that perform Music Emotion Recognition (MER) on the brain signals. No matter the use case, this

section concentrates on feature extraction methods used and on what ML approach the researchers used for their goal.

Following Chapter 3, Chapter 4 discusses different music datasets that can be used for the purposes of this study. At first, the main requirements for a music dataset being used in this study are explained and clarified. Based on these requirements, five different song databases are presented and evaluated against the mentioned requirements. This chapter then concludes with the final decision on which dataset fits the terms best, thus being used in the study.

With the previous four chapters laying the groundwork, Chapter 5 illustrates the whole process of the study from start to finish. At the beginning, the acquisition of data is emphasized. Before talking about the EEG system used for measurements, the general experiment setup and environment are presented alongside the stimuli in each session. Furthermore, the technical setup is shortly explained by highlighting how each component works together. This section is then followed by a discussion of the denoising steps made on the raw signal, the artifact removal strategy, and the general preprocessing pipeline. Having preprocessed data, the following section deals with the features extracted from the signal. Similar to Chapter 2, each domain is covered, showcasing which features were extracted in which domain. Last but not least, the approach regarding classification is shown. After discussing the general validation and hyperparameter tuning approach, this is further specified by each individual ML used for this project.

Chapter 6 then discusses the results also in regard to the goals of this study. This chapter evaluates the performance of each classification model based on predefined metrics.

Last but not least, Chapter 7 concludes the research project as a whole. It not only sums up the thesis's key findings but also mentions problems and limitations, proposing future research on the basis of this study.

2 Theoretical Background

2.1 Medical Foundations of EEG Signals

Before talking about how to EEG signals can be used for classification or further analysis, it is vital to understand how these signals can be measured non-invasively and how they depict the brain's activity.

The Central Nervous system (CNS) of nerve and glia cells can be found between neurons. Nerve cells can recognize and transmit stimuli through other nerve cells to further locations. To understand how this is possible, the structure and functionality of a single nerve cell have to be understood. This kind of cell consists of three main components. Firstly, there is the cell body with a single nucleus. It is responsible for the cell's metabolism and protein synthesis. Secondly, the Axon, a long cylinder, transmits electrical impulses up to a metre through the body. The third component, dendrites, further supports the transportation of electricity. They are either connected to other dendrites or axons of neighboring cells from which they can receive impulses. Vice versa, it is also possible to transmit the impulse to another cell [27].

Related to the transmission of electrical impulses, the so-called Action potential (AP) is very important regarding EEG, since the AP represents information sent by a nerve. In simple words, AP is the temporary change in the membrane potential that is transmitted along the axon caused by the exchange of ions. It lasts between 5 and 10 ms, resulting in a peak during the positive polarization of the membrane. After repolarizing to a negative potential, the nerve returns to resting potential. It is important to remember that an AP is only generated if there is a strong enough stimulus. If the stimulus is not powerful enough, only a small local electrical impulse is created, which is not enough to be transmitted to another nerve, thus not resulting in an AP [27].

As already mentioned, the AP is a key player when it comes to EEG because an EEG signal indirectly records current flows during the excitation of the synapses. During the transmission of an electric impulse because of the activation of neurons, an electrical field is created next to a magnetic field, which can be measured by specialized EEG systems over the scalp (cf. Section 2.3). Since the brain and the scalp consist of multiple layers, only a large population of active neurons can be recognized by systems measuring from the scalp hence they have to be amplified for visual inspection [27].

The measured signal can differ based on the patient's mental or emotional state.

Therefore, one has to try to distinguish the signal into components to make analysis easier.

2.2 Brain Rhythms

Visual inspection is traditionally the way to do when analysing EEG signals for brain disorders or other malfunctions. Specific amplitudes and frequencies of these signals change in a healthy human depending on their state. For instance, the EEG signal differs between wakefulness and sleep. To distinguish the different appearances of the signal, the frequency range is taken into account to form five brain rhythms (cf. Figure 2.1), each of which defining a particular state of the brain [27].

Delta waves occupy the range of 0.5-4 Hz and represent the brain's grey matter [20]. They mainly occur in deep sleep and may be present in the waking state. These signals are very easy to confuse with large muscle artifacts from the jaw or neck, since they are close to the skin's surface. Additionally, the Delta signals coming from the brain are highly attenuated in passing through the skull [27].

Theta waves lie between 4-8 Hz and are primarily present in the thalamic region. They appear when a human's mental state transitions from consciousness towards drowsiness, which is why they dominate during meditation therapy. [27, 20] Combined with that, it also stands for human growth as they are linked to serotonin hormones [20].

Within the range of 8-13 Hz, Alpha signals are present and are shown as a round-shaped signal. Being the most dominant rhythm in the whole brain and standing for the brain's white matter, it indicates relaxed awareness without any attention and concentration [27, 20]. Studies also show that Alpha waves are a waiting pattern by the visual region of the brain since they are reduced or even eliminated by opening the eyes [27].

Signals with frequencies between 14-26 Hz [27] or 13-30 Hz [20] are classified as Beta waves, which stand for the usual waking state of the brain. Therefore, it is associated with active thinking or problem solving [27]. By inducing the production of Cortisol, it is responsible for learning and memory [20].

Everything above 30 Hz (up to 45 Hz) is considered as Gamma waves. They are very rare in occurrence and are primarily used for detecting and confirming brain diseases [27]. However, they also appear during hyper alertness and the integration of sensory inputs, which is why they combine senses and memory [20].

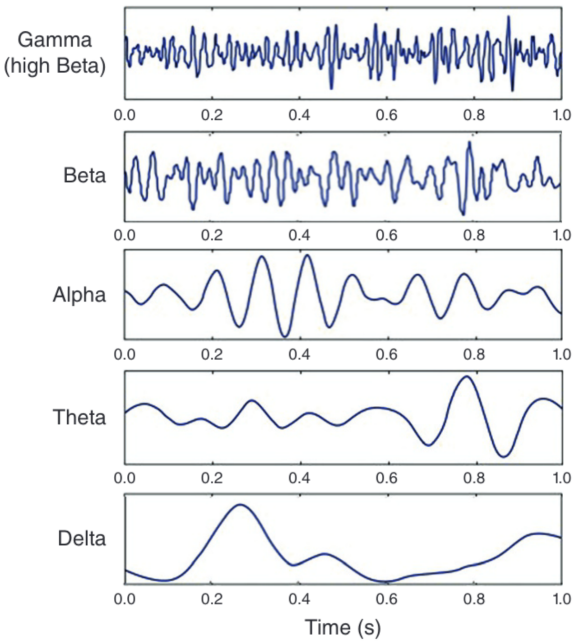


Figure 2.1: The five brain rhythms in EEG [27]

2.3 Recording and Measurement of EEG Signals

In the previous two sections, it became clear that the electrical activity coming from APs can be used to depict the neural state of the brain. Therefore, covering how these impulses can be measured is also important.

An EEG system mainly consists of multiple electrodes (called channels), differential amplifiers for each channel, and filters. Multichannel Analogue to Digital converter (ADC) are employed as well to convert the measured signal points to enable the connection to computerized systems [27].

The most crucial components for data quality are the electrodes. There are different types of electrodes depending on the study's use case (disposable, reusable disc electrodes, headbands, etc.). However, headbands with Ag-AgCl discs are the most popular for multichannel measurements. As already mentioned in Section 2.1, because of the layered anatomy of the scalp, there is high impedance between the cortex and the electrodes, which can lead to deformations in the signal. Therefore, a rule of thumb is that the impedances should be less than 5 $k\Omega$ and balanced to 1 $k\Omega$ within each other [27].

For unification in measurement and placement in multichannel EEG recordings,

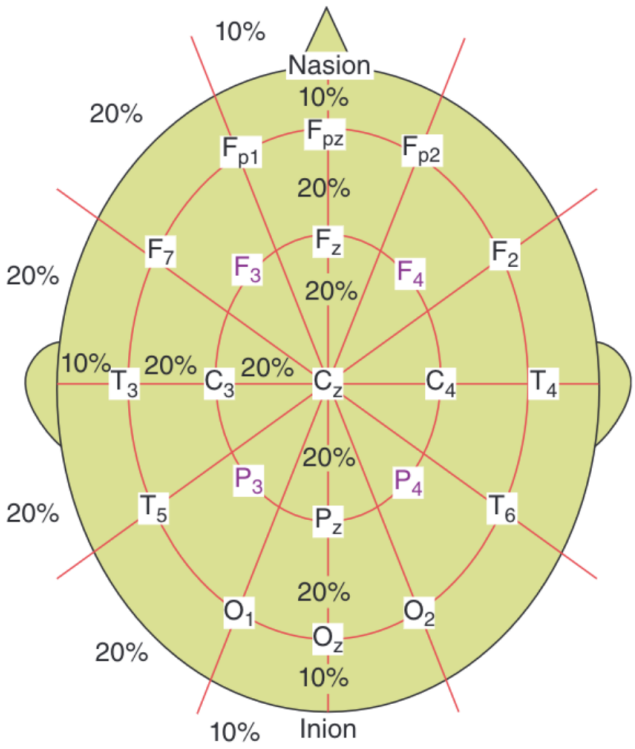


Figure 2.2: Conventional 10-20 EEG electrode positions for 21 electrodes [27]

the 10-20 system is recommended as a conventional electrode setting. A setup for 21 electrodes is illustrated in Figure 2.2. This system avoids eyeball placement and employs constant distances between all electrodes. It is called the 10-20 system because the electrodes are placed at intervals of 10 or 20% of the total front-back/left-right distance of the skull. In the case where more than 21 electrodes are employed, the additional channels would be placed between the existing electrodes with equidistance between each other [27].

As the adequate bandwidth of EEG signals lies around 100 Hz, the minimum sample rate should be 200 Hz to meet the Nyquist criterion[1] Depending on the required resolution, sampling frequencies up to 2000 Hz are possible. In quantification of the signal, 16 bits for each sample is sufficient and popular for many EEG recording systems [27].

[1]The Nyquist criterion states that to accurately reconstruct a signal without aliasing, it must be sampled at a rate at least twice its highest frequency component.

2.4 Denoising and Preprocessing of EEG signals

After acquiring the EEG signals, further steps have to be followed before performing a classification task. Since EEG is a non-invasive technique, other signals, so-called artifacts, are also measured. Different artifacts are presented before discussing possible methods to remove the data.

2.4.1 Types of Artifacts in EEG signals

Physiological artifacts	**Non-Physiological artifacts**
Ocular-related artifacts: eye movements and blinking	Power line interference
Electromyography (EMG) artifacts in head movements	Poor placement of electrode on scalp
Electrocardiography (ECG) artifacts in heart beat	Malfunction of a part in the EEG recording system
Scalp perspiration and movements	Digital artifacts due to loose wiring/-circuit board connections

Table 2.1: Types of artifacts in EEG [17]

Before illustrating what types of artifacts exist, it is essential to mention that artifacts themselves do not destroy EEG signals but add to the neural activity. Therefore, the types and characteristics must be explained thoroughly before removing any of this data. In Table 2.1, an overview of forms of artifacts can be found. In general, artifacts can be put into two groups: the *Non-Physiological artifacts* and the *Physiological artifacts*. As the name already suggests, while non-physiological artifacts are distortions from the outside world, physiological artifacts are interferences by the human body. [17] These groups will be described in more detail in the next two sections.

Physiological Artifacts

As mentioned, Physiological artifacts are characteristics of the human body, including ocular movements, head movements, heartbeats, and muscle noises. For most of these artifacts, it is beneficial to record their corresponding Biosignals to be able to retrace them in the measured signal. [31]

Ocular artifacts can be measured by Electrooculography (EOG) and are mainly produced by eye movements and blinking. The latter generally produces a much

larger amplitude than the EEG activity. Because of the anatomy of a human, EOG contaminations mostly occur in the frontal electrodes. [31]

An even more common artifact comes in muscular activity, which can be measured by an EMG signal. Appearing in studies where the person is awake, the main actors for these kinds of contaminations are swallowing, talking, and walking, depending on the experimental environment. Most cranial EMG signals are in charge of disadvantageous effects in the recorded signal. However, it is difficult to detect and correct them as EMG not only represents a wide range on the spectral distribution, but also does not repeat, making it even more difficult to classify them. [31]

Last but not least, the activity of the heart (ECG) plays a significant role in contamination of measured EEG signals. In contrast to EMG data, a regular pattern is visible, which can be further classified by comparing it to ECG measurements. This way, it is relatively easy to remove those from the signal. [31]

Non-Physiological Artifacts

On the other hand, the so-called non-physiological artifacts are coming from outside-world interference. The most common source is electrical interference from power lines (50 Hz in Europe). This generally can be reduced by shielding the cable or moving the subject far away from the power source. [17]

Another very common non-physiological artifact is the poor placement of electrodes on the scalp as the created DC potential is disturbed. Therefore, electrodes should be checked regularly to ensure high quality recordings. [17]

Another obvious source for those kind of artifacts is the malfunction of any part within the measurement system. For example, amplifiers can create noise caused by the thermal agitation of electrons within the circuit since their are small and have a small single chip. [17]

Last but not least, digitial artifacts are artifacts originating from digitizing, storing, or processing the EEG signal in a computerized system. The main reason for that is loose wiring, leading to mistakes in the analog to digital conversion and skew errors. [17]

2.4.2 Artifact Removal Techniques

Before presenting state-of-the-art techniques for removing artifacts, it is essential to mention that most of these algorithms and techniques concern physiological artifacts. Controlling the recording environment can correct or altogether avoid most non-physiological artifacts. By checking the placement of the patient and the electrodes on the scalp, non-physiological artifacts remain a small problem in artifact removal compared to handling physiological artifacts. [17, 31]

The first technique is the so-called (Frequency) filtering. It not only removes the power line interference but also removes high- and low-frequency noises. Depending on the goal of the filtering, different filtering techniques are available, mainly a low-pass filter, a high-pass filter, a band-pass filter, and a band-stop filter. While a low-pass filter removes all frequencies above a chosen threshold, a high-pass filter eliminates frequencies below a boundary. The implementation of band-pass filters requires the definition of a lower and upper frequency boundary. When applying, it removes all frequencies below the lower frequency and above the higher frequency (so keeping all frequencies within the given constraints). Last but not least, a band-stop filter does the opposite. It removes all frequencies within the chosen interval, thus keeping the frequencies below the lower and above the upper bounds. Frequency filtering should be the first step of preprocessing EEG signals (even before segmentation) to remove low- and high-frequency noise outside the spectrum of brain rhythms. [17]

When discussing algorithms and techniques dedicated to removing artifacts, assumptions about the recorded signal must be met. The most important assumption is that the recorded signal is the sum of the brain activity and noise. A more generalized version describes the measured EEG signals as linear mixtures of waveforms descending from multiple brain sources and artifacts, ultimately propagating to the scalp. Mathematically, this can be described by the following mathematical equation:

$$\mathbf{X} = \mathbf{AS} + \mathbf{V}, \tag{2.1}$$

where $\mathbf{X} \in \mathbb{R}^{n \times N}$, $\mathbf{A} \in \mathbb{R}^{n \times m}$, $\mathbf{S} \in \mathbb{R}^{m \times N}$,$\mathbf{V} \in \mathbb{R}^{n \times N}$. While N represents the total count of samples measured, n stands for the number of EEG channels, and m for the number of total sources the signal originates from. In Equation 2.1, $\mathbf{X}$ describes the EEG signals resulting from the measurement. While $\mathbf{V}$ stands for added noise[2], $\mathbf{A}$ is an unknown mixing matrix and $\mathbf{S}$ is a matrix of unknown Sources representing brain sources and artifacts. [31]

With this model, the goal is now to solve the Blind Source Seperation (BSS) problem, so in this case, separating neural activity from artifact activity. For this problem, component-based methods are used because they find principal or independent components equal to the input EEG channels. After processing in the transformed domain, an inverse transformation is applied with the corrected components, resulting in processed EEG data minimising general noise and maximising the overall neural information in the signal. [31]

In general, BSS aims to estimate $\mathbf{S}$ based on $\mathbf{X}$ in Equation 2.1. It must be mentioned that those methods' effectiveness lies in different assumptions that have to be met to a certain extent, e .g., independence or non-Gaussianity. The more fulfilled these

[2]In the following, it is assumed that $\mathbf{V} = 0$

assumptions are, the more effective the BSS method is. Apart from that, the success of such methods also depends on correctly identifying the sources as either brain activity or artifacts, which can serve as a challenge depending on the extent of artifacts. [31]

One of those methods is Principal Component Analysis (PCA), which uses an orthogonal transformation to turn the observed data to linearly uncorrelated variables (Principal components). The resulting components have the property of having the most significant possible variances while being orthogonal to each other. However, current research suggests that artifacts and cerebral activity can be modeled better as independent instead of orthogonal. To use this method, Independent Component Analysis (ICA) has emerged as the leading method for BSS. [31]

ICA consists of multiple related methods, separating linearly mixed signals by taking advantage of the statistical independence of the sources. The algorithm usually works based on Higher-order statistics (HOS). After prewhitening the covariance matrix of the data, ICA tries to find a linear transformation $\mathbf{W}$ for estimated sources $\hat{\mathbf{S}} = \mathbf{WX}$. $\hat{\mathbf{S}}$ is then supposed to be as independent as possible. How effective the ICA depends on the independence of sources and the full column rank of the mixing matrix $\mathbf{A}$. Since this algorithm relies on statistical features, the number of samples must be sufficiently large. In most cases, it is recommended to use all available data because cerebral activity and artifacts are spatially stationary through time. [31]

2.5 Feature Extraction of EEG Signals

Before classification tasks can be performed on the denoised signal, it is essential to map the characteristics of each signal to specific metrics, called features. In biomedical signal analysis, big data has become more common, thus leading to high-dimensional sample data. Therefore, the biggest goal of feature extraction is to reduce this dimensionality and represent the data more compactly. Overall, feature extraction techniques not only allow for the reduction of complexity but also for saving on resources and computational time [28].

Different ways of extracting features from the EEG signal exist. Depending on the task and circumstances, distinct measures can lead to independent and different results. Therefore, it is important to explore as many different features across multiple domains as possible. Mainly, there are three domains from which features can be derived from, namely the time domain, frequency domain, and time-frequency domain, each extracting features based on different attributes of the EEG signal, capturing vestatile characteristics of each EEG channel. For this reason, it is essential to further perform Feature selection techniques like the Minimum Redundancy Maximum Relevance (MRMR) selection technique to keep the dimensionality low and work with an optimal

set of features. The following sections explain characteristics of the domain's features alongside examples, which is then concluded by presenting the MRMR feature selection algorithm.

2.5.1 Time Domain Feature Extraction

In time domain feature extraction, signals are analysed to quantify how they change over time. Depending on the number of data points in a session, windowing is employed to achieve desirable results [28].

One of the simplest and common approaches is calculating basic statistical features from the signal at hand. This includes mean, standard deviation, maximum, minimum, median, variance, range, skewness, and kurtosis [28].

In addition to these basic statistical features, parameters tailored for signals can be used to achieve desirable results and information. One of those is Root Mean Square (RMS), which measures the average power of a signal, representing a signal's energy or loudness. It can be calculated with Equation 2.2, where N is the number of samples and $V(i)$ the amplitude of the signal of the i-th sample [25].

$$V_{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} V^2(i)} \tag{2.2}$$

To depict the noisiness of the EEG signal, the Zero Crossing Rate (ZCR) can be measured. As the name suggests, it counts the number of sign changes and normalizes the value with the number of samples.

In information theory, the (Shannon) entropy describes a signal's uncertainty and information content in bits. A low entropy means the signal is predictable, tending to a constant signal, whereas a high entropy represents a trend towards random noise [4]. The Shannon Entropy can be computed using Equation 2.3. p_i hereby stands for the probability of the i-th data point within the sample window [13].

$$H = -\sum_{i=1}^{n} p_i \log_2(p_i) \tag{2.3}$$

Additionally, used in recent EEG studies, the so-called Hjorth parameters gained more popularity as they are able to quantify power, frequency content, and signal shape variation. These parameters are three features, the activity (Equation 2.4), mobility (Equation 2.5) and complexity (Equation 2.6) of a signal $x(t)$.

$$h_0(x(t)) = var(x(t)) \tag{2.4}$$

$$h_1(x(t)) = \sqrt{\frac{h_0(\frac{dx(t)}{dt})}{h_0(x(t))}} \tag{2.5}$$

$$h_2 x(t) = \frac{h_1(\frac{dx(t)}{dt})}{h_1(x(t))} \tag{2.6}$$

Mathematically, each parameter stands for a specific characteristic of the signal. While the activity stands for the signal power, the mobility depicts the mean frequency and the complexity of the change in frequency [32].

2.5.2 Frequency Domain Feature Extraction

The frequency domain is an analytical space in which the EEG signal is interpreted in terms of frequency [8]. While in the time domain, the data is conveyed as a signal amplitude change, the frequency domain represents the change of its signal power. To do that, the signal has to be transformed from the time to the frequency domain by spectral estimation. This technique estimates the Power Spectral Density (PSD) of a function to detect periodicities in the data. Therefore, one could say that the signal is decomposed into distinct sinusoids, which is further illustrated in Figure 2.3. In this example, the signal can be described as three sinusoids with 2 Hz, 8 Hz, and 20 Hz and amplitudes of 1, 0.8, and 0.5, respectively [17].

One of the most accurate methods to compute PSD is Welch's method based on a Discrete Fourier Transform (DFT). In general, DFT creates the representation of the signal in the frequency domain. A Fourier transform characterizes a time-dependent signal as the summation of multiple sines and cosines. Welch's method now divides the signal into K windowed data segments of M samples (overlapping by D samples) and computes the DFT over each segment. The average of all results then leads to the final spectral estimation. This approach causes a reduction in variance by $1/K$, making it more precise than other standard methods [17].

Multiple different features can be extracted from the PSD to access the frequency-related characteristics of an EEG signal. One of the most prominent EEG spectral features is hereby the band-limited power. It represents the spectral power within a particular frequency band, i. e., within the frequency range of each brain rhythm. It can be calculated by making use of Equation 2.7. The sum of all power values P at each frequency point f_i within a frequency band I_{band}. For normalization and elimination of individual differences, the relative power can be taken by dividing by the total power in the whole frequency range [17]. This method allows a detailed analysis of the different mental states during a session.

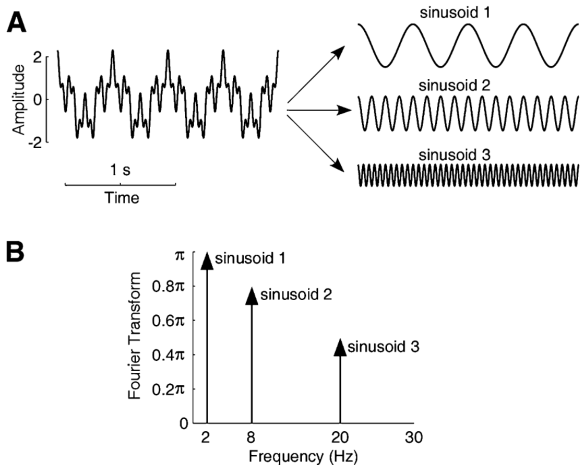


Figure 2.3: Comparision between Time and Frequency domain. From the signal (on the right), sinusoids (on the left) are extracted that have unique frequencies and amplitudes (at the bottom) [17]

$$P_{band} = \frac{1}{N} \sum_{f_i \in I_{band}} P(f_i) \tag{2.7}$$

Furthermore, the spectrum can be treated as a random process, thus calculating statistical features similar to those mentioned in the time-domain (cf. Subsection 2.5.1) [17].

2.5.3 Time-Frequency Domain Feature Extraction

Many useful features can be computed from the frequency domain. However, it is assumed that the spectrum of an EEG signal does not change over time. This assumption oversimplifies the signal at hand since EEG is defined by non-stationary and dynamic behaviours, leading to a change of the spectrum over time [17].

This problem can be solved by combining the time-domain with the frequency-domain; Time-frequency Analysis (TFA) methods emerged to simultaneously analyse time and frequency characteristics. There are two main types of TFA techniques. The first one comprises methods that assign signal power per frequency point. These so-called Time-frequency power distributions can be computed by an Short Time Fourier Transform (STFT) or a Continous Wavelet Transform (CWT). The second class of techniques is called Time-frequency signal decomposition. As the name suggests, it decomposes the signal into additive components concerning their Time-frequency

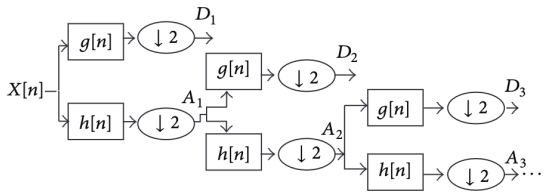


Figure 2.4: Visualization of a DWT [10]

properties. In that case, methods like Discrete Wavelet Transform (DWT) or matching pursuit can be employed [17]. In the following, Wavelet Transform techniques will be explained.

Generally, Wavelet transform methods compress a time-varying signal into a few parameters representing the signal, called wavelets. At the core, wavelets are localized, small wave-like functions that refer to the different positions and scales of the signal. A Wavelet transform can analyse various data characteristics depending on the window size. While long windows extract low-frequency components, short windows get high-frequency information. This windowing technique enables capturing not only coarse but also fine details [10].

A mother wavelet function ψ is used to generate the single wavelets by shifting and scaling ψ. To achieve this, two parameters play a key role. Firstly, the scale parameter a is responsible for determining the wavelet's resolution. Secondly, the translation factor b controls the shift of the time domain. In general, there are two types of Wavelet transform methods: the CWT and the DWT.

To compute the CWT coefficients of a signal $x(t)$, following equation is used:

$$CWT(a,b) = \int x(t)\psi^*_{a,b}(t)dt \tag{2.8}$$

The signal is probed over every defined combination of a and b to calculate a 2D-matrix, where each entry corresponds to a specific coefficient [10].

One of the biggest problems of CWT is the result of a lot of unused information and computational effort due to the continuous changes in a and b. To solve this issue, DWT can be employed instead, which is showcased in Figure 2.4. Each step (or level) consists of two digital filters $g[n]$ and $h[n]$ and two downsamplers by two. $g[n]$ stands for the mother wavelet function and acts as a high-pass filter, while $h[n]$ is its mirror function working as a low-pass filter. Each stage outputs a detail of the signal D and an estimation of the signal A, which is then used as an input for the next step. This results in a hierarchical set of coefficients, which become more accurate the higher the level is set [10].

2.5.4 Feature Selection

Based on the previous sections, it becomes clear that many different kinds of features can be calculated based on an EEG signal. The number of features, especially scales, in EEG based classification problems, since all features are computed for each channel and the metrics are very versatile. However, depending on the dataset, many features can lead to undesirable results, such as Over- or Underfitting. Therefore, it is vital to employ standard feature selection methods that choose the optimal subset of all these features. This not only leads to a reduction in dimensionality but also to choosing those features that hold the most information. One feature selection method in that domain is the MRMR feature selection, which will be explained in the following.

As the name suggests, the MRMR algorithm tries to find the feature set that has maximal relevance and minimal redundancy between each other. On the one hand, relevance V describes how much information a feature set F provides about the different class labels. On the other hand, redundancy W defines how much information is shared between features in the feature set [18].

With both W and V at hand, the goal of the MRMR algorithm is to maximize W while minimizing V. The mutual information quotient $MIQ = \frac{V}{W}$ or the mutual information difference $MID = V - W$ can be considered for this. To find the optimal feature subset S with size M, features $f \in F$ are iteratively added to S to maximize the criterion. However, this method can be computationally expensive since it has an exponential runtime depending on $|F|$ and M [18].

3 Related work

3.1 Music Genre classification with audio features

Before talking about how one can derive music genres from EEG signals, it is important to briefly mention research concerning MGR based on the song itself. This serves as a proof of concept that audio features differ between different genres. To do that, recent research is presented in this section, focusing on features and ML models used.

In 2018, Bahuleyan compared the performance of Deep Learning (DL) models with different ML models in labeling audio signals with their predefined genres. While the DL models were solely trained with the spectrogram of each song, the ML models are based on manual feature extraction in the time and frequency domains. The songs come from the Audio Set[1], consisting of seven genres, including Pop, Rock, Hip Hop, Techno, Blues, Vocal, and Reggae. The performance was measured across accuracy, F-score, and AUC. Hereby, the Convolutional Neural Network (CNN) has performed best with an accuracy of 64%. Under the ML models, Extreme Gradient Boosting (XGB) performed best amongst Logistic Regression, Random Forest, and Support Vector Machine (SVM) with an accuracy of 59%. This shows that standalone DL models perform better than standalone ML models. However, further evaluation has demonstrated that an ensemble classifier consisting of the best performing CNN and XGB performs best overall with slightly better evaluation scores than the fine-tuned CNN [3].

Similar to the previous study, Haggblade et al. investigated different ML algorithms to classify songs between four different genres: classical, jazz, metal, and pop. As a source of songs, they used the GTZAN dataset[2] and filtered all the audio for the four most distinct genres. As features, Mel Frequency Cepstral Coefficients (MFCC) are the only extracted attributes. With a 70 to 30 train-test split, various ML models (including SVM, neural networks, k-means, and k-NN). It is important to mention that this study employs a Directed acyclic graph (DAG) SVM, where each node represents a two-class SVM trained on each combination of genres. A process of elimination is then used to decide the final classification of the multi-class classifier. When evaluating these models regarding accuracy, the results varied between the algorithms. Although the accuracy also depends on the genre (e.g., most ML models have difficulty classifying Jazz), the

[1] cf. Section 4.1
[2] cf. Section 4.5

researchers found that k-NN and k-means had similar performance of 80%. The DAG SVM reached an accuracy score of 87%. The neural network yielded an accuracy of 96%. According to the study, this enormous gap between SVM and Neural Networks is surprising, as they expected similar performance. [15]

In conclusion, both research papers have proved that the differences in genres are learnable by ML and DL models, showcasing desirable performance in multi-class classification. However, both studies have found that Neural networks perform better than traditional ML algorithms since they do not rely on manual feature extraction [3, 15].

From the perspective of the thesis and the goal of classifying music genres based on EEG signals, the fact that ML techniques can identify differences across genres supports the fact that the neural response should be different from genre to genre, which can be captured in the EEG data. These differences and patterns can then be identified by the ML model in theory.

3.2 EEG datasets and studies with music stimulus

Besides DEAP, I mention two other datasets, if time is there, swap them and write about them in here instead of the ones i already have there. This would make the whole Related work section more cohesive and easier to follow. Papers are linked in the single papers for further reading. I can remove this by saying at the beginning of each dataset who used the particular datasets. However, ask Simon abt it once more

As music is a popular stimulus used in EEG studies to collect the signal data, many research projects have been conducted in this area. This section intends to serve as an overview of work that has been done in the context of constructing an EEG dataset where music served as a stimulus. For each dataset, the experiment setup and acquisition methods are described, and certain preprocessing and denoising steps are emphasised.

3.2.1 DEAP

The Database for Emotion Analysis using Physiological Signals (DEAP) is an EEG dataset made public in 2012 by Koelstra et al. It is a multimodal dataset providing data for analysing human affective states. For 32 participants, EEG and peripheral physiological signals were recorded while watching 40 one-minute sections of music videos. The video was also shown alongside the music, serving as an additional stimulus. After each snippet, test subjects were asked to rate each music video in terms of valence (positively or negatively connoted), arousal, liking, dominance and familiarity. [19]

The music comes from a website called Last.fm and was selected using a sophisticated procedure. 120 music videos were initially selected and labelled with keywords out of a list of emotional keywords provided by the Last.fm website. Based on each emotional keyword, ten songs labelled most with one of them have been selected in the next step of the procedure. The chosen songs were further filtered according to valence and arousal attributes. After this filtering step, a one-minute highlight section was selected with an effective highlighting algorithm from each music video. Last, a web-based assessment experiment was conducted to decide the 40 final music video clips for the DEAP study. [19]

The experiment was conducted in a laboratory environment with controlled illumination. The EEG signals have been recorded with a sampling rate of 512Hz by 32 active AgCl electrodes placed in the international 10-20 system. Apart from that, not only have physiological signals been measured, but frontal face videos for 22 of the 32 participants have also been recorded. The basic procedure of each session includes a five-second baseline recording before displaying the one-minute music video. When the video ended, the participant assessed the clip based on arousal, valence, liking, familiarity and dominance. [19]

For further investigation of the recorded signal data, the researchers provide a preprocessed variant of the EEG data. First and foremost, the data were commonly referenced and then downsampled to 256Hz. The downsampled signal went through a high-pass filter with a 2 Hz cutoff frequency employing the EEGlab toolbox. Eye artefacts were removed by using a BSS technique. For further analysis, the baseline recording of five seconds was removed. [19]

3.2.2 Music BCI

Behavioural responses like in the DEAP dataset can be researched, as well as the perception of musical patterns. Therefore, researchers from Berlin and Copenhagen have chosen polyphonic music as a stimulus for an EEG dataset. Polyphonic music describes music with two or more parallel melodic lines, including different instruments. In their work, the main goal is to explore the suitability of polyphonic music in Brain-Computer Interfacing (BCI). [29]

For this purpose, eleven participants aged 21-50 years were chosen, each not educated in the field of BCI research. For acquisition, a 1000 Hz sampling rate was determined with 64 electrodes placed according to the international 10-10 system. The stimuli were 40-second music clips, each of which contained six tracks. Two of these tracks are meant for stereo audio playback, the other four are a mono mix of the stereo clip recorded as additional EEG channels. Each clip contains three instruments playing musical patterns, where one instrument plays a deviant version while the other two

stay the same. Per instrument, there are 3-7 deviations per clip. Synth-pop, which has around 130 bpm, and jazz, which has around 120 bpm, were chosen as genres. While the Synth-Pop clips consist of drums, a bass, and a keyboard, the clips coming from Jazz comprise flute, piano, and a double bass. Ten different clips were made for both music types with those kinds of deviant patterns. [29]

During the experiment, the participant was seated comfortably in a chair and instructed to stay still and relaxed, especially to minimise eye movements during the trial. The main experiment is divided into ten blocks with 21 music clips played randomly. The listener was told to pay attention to each of the three instruments in seven different music clips. To further ensure attention, the participant has the task of counting the number of deviants in the cued instrument. Before the following clip, the participant must enter the recognised number of deviations. [29]

The signal data were preprocessed to further analyse the recorded EEG data. After downsampling them to 250 Hz, they were low-pass filtered using a Chebyshev filter accompanied by a passband of 42 Hz and a stopband of 49 Hz. The data is sectioned into epochs from -200 ms to 1200 ms for additional denoising. While the prestimulus interval was used for baseline correction, a min-max criterion was employed to discard physiological artefacts. [29]

3.2.3 MUSIN-G

In 2022, researchers published an open-source EEG dataset examining the correlation between the brain and music genres. In the experiment, 20 participants listened to naturalistic songs (meaning mostly without lyrics) across 12 genres, ranging from Indian folk music to Goth Rock or Western electronic music. Apart from the genre, subjective characteristics like familiarity and liking have been considered [24]

It is essential to mention that all participants have little musical training and are therefore considered non-musicians. The participant's EEG data were recorded by a 128-channel high-density EEG cap while sitting comfortably in a dark environment. The main procedure can be divided into two parts. At the start of the acquisition, a baseline recording was conducted during absolute silence lasting for two minutes. During this process, the subject's eyes were closed. The second part then consists of listening to the songs, where the order is randomised depending on the person hearing the audio. A beep sound instructed the participants to close their eyes when the song started to play. After each song, a ten-second gap was chosen, followed by another beep sound indicating to open their eyes for answering two questions. The participant has to rate the song on a scale of one to five regarding familiarity and liking. After answering the questions, the next song was played with a trailing ten-second gap in between. [24]

Since the data is provided as raw and segmented EEG data, the signals can be

preprocessed for further analysis. The authors suggest a workflow that includes the EEGLAB toolbox in MATLAB. First, the signals are high-pass linear FIR filtered at 0.2 Hz. This is followed by the CleanLine method, which removes 50Hz of line noise (comparable to a notch filter). Before segmenting the data into each song, the filtered data were down-sampled to 250Hz. By employing a spectrum criterion, bad channels were removed to prepare them for the artefact removal. The Multiple Artifact Rejection Algorithm (MARA) was used to remove the artefacts, an ML-based method for artefact detection and removal. Lastly, removed channels are interpolated with the spherical function of EEGLAB before the average reference is applied to the whole dataset. [24]

3.3 Music Genre classification with EEG signals

Overwork this as well after doing some more research, similar to Emotion Related work

With the help of EEG datasets similar to the ones described above, researchers can use the acquired data to perform classification tasks in affective computing. Using personal data like brain signals can capture content-centric features like personal preference, which would not be possible with audio features [12]. Most of this research done in this field deals with predicting a person's emotional state, which will be covered in Section 3.4. Although not many studies regarding MGR based on EEG signals have been published, a recent study is presented in the following.

In 2016, Ghamemaghami and Sebe studied whether performing MGR based on brain-recorded signals is possible. As a source of EEG data, the authors employed the preprocessed version of the DEAP dataset[3]. Apart from EEG signals, the researchers also took Magentoencephalography (MEG)[4] signals into account by employing the DECAF dataset, which holds MEG data from 30 participants watching the same clips as in the DEAP study. Because neither dataset provides labels for genres, all 40 song clips were manually annotated in two distinct categories, Pop and Rock. By doing this, the classification task is a simple binary decision between either of the two tags [12].

Before performing the classification, features have to be extracted. For both EEG and MEG, the averaged power over each of the four brain rhythms (alpha, beta, theta, gamma) was taken into account. Furthermore, multimedia content features were calculated from the song's audio and music video as a comparison. An SVM classifier was used with a leave-on-clip-out-cross validation technique, measuring accuracy and F-score. It was classified not only on the level of subjects (i.e., only using the brain data from each subject) but also on the whole population (i.e., taking all subjects into

[3] cf. Subsection 3.2.1

[4] Non-invasive neuroimaging technique measuring the magnetic fields produced by electrical activity from the brain

account). On the subject-level, the researchers achieved a better accuracy than chance level when only employing EEG signals, which also shows a significant difference. When fusing the EEG signal with the multimedia features, the classifier provides an even higher accuracy compared to only using EEG signals [12].

For population-level analysis, the subject-level descriptors were considered to calculate the genre over a majority-vote, outperforming both the accuracies of subject-level evaluation [12].

This research shows the first evidence that brain signals like EEG can be used for classifying music genres, demonstrating the potential for user-centric music retrieval. Furthermore, since the fusion of brain signals with multimedia content has shown significantly better results, it suggests that there is complementary information between both [12].

3.4 Emotion classification with EEG signals

Since work in EEG based MGR is limited, it is vital to investigate further research on classification tasks related to EEG signals. As listening to music is highly linked with the emotional state of the human brain, it is possible to use extract patterns out of a measured EEG signal to map them to different emotional parameters like valence or arousal. As those reactions are highly correlated with the music's genre, it is important to get an overview of recent studies regarding Emotion recognition using neural activity.

TBD

As a basis, it is important to clarify where the EEG data orignates from that is used for classification. This is essential since the results and performance of classification tasks heavily depends on the EEG signal. To ensure reproducibility and comparision between different work, most studies employed publicly available EEG datasets that are specifically designed for emotion tasks. Very prominentely used is the DEAP[5], since it is a dataset that already offers preprocessed data sepcified on emotion recognition tasks [22, 16, 2, 6].

Some studies also combine the DEAP dataset with the SJTU Emotion EEG Dataset (SEED) for evaluating their results across different environemnts and stimuli. In this case, film clips were chosen as an emotional excitement. The EEG signal is mesured by a 62-channel system labelled with either positive, negative or neutral state [22, 16].

Apart from that, the Muse headband EEG dataset was utilized by Chatterjee and Byun in 2020 and Abdulamikov et al. in 2024. In this dataset, two subjects watched 60 second clips from movies designed to arouse positive or negative emotions. The EEG

[5] cf. Subsection 3.2.1

signal measured by four channels is also available after applying preprocessing steps and labelling similar to the SEED data [7, 1].

To explore the effectiveness of different stimuli and to fully control the environment, Lin et al. performed their own experiment involving simple music listenting and evaluation over the 2D valence-arousal emotion model, mapped to four basic emotions (joy, anger, sadness and pleasure) [23].

Besides the EEG signal itself, feature engineering is crucial for the performance and accuracy of classification. As EEG signals are very versatile, many different features can be extracted from the neural activity. In various related research, parameters have been extracted from the time-domain capturing time-dependent attributes of the signal. Besides classic statistical features like mean, standard deviation, etc. [7, 33], the most used temporal features included (differential) Entropy and Higuchi's Fractal dimensions [2, 16].

Furhtermore, most of the studies that perform classification on EEG data use frequency related features as it is possible to extract information from certain frequency bands. In this case, these frequency bands can be defined for each brain rhythm, hence being one of the most precise measures for brain activity. It is therefore commonly used across studies related to EEG.

4 Music Datasets

Since this thesis deals with music genre classification, a suitable set of musical stimuli has to be chosen to gather valuable data. As research in the domain of music is widespread, multiple music datasets exist that are used in different studies. Therefore, the difficulty lies in choosing the right dataset for this thesis. However, the work goal crystallises precise requirements for such a music dataset.

First of all, the songs that are listened to are supposed to be around 30 seconds long. The reason is that the stimulus could be lost if the song runs too long because the brain can adapt to the genre-specific features, thus reducing the visibility of the impact on the brain. Apart from that, the number of songs should be large, with around 800 to 1000 song clips. A large pool of songs is essential to get as much different and versatile data as possible. This helps in future steps regarding classification. However, the genres used in this pool must also meet specifications related to this requirement. First, the songs must be clearly labelled to avoid confusion in later stages of the study. It is also essential to know how balanced the dataset is in terms of genres listened to. This is supposed to be as balanced as possible so that each genre is represented equally. Additionally, it should be easy to employ this dataset in the study. It should be uncomplicated to load the songs and get access to labels or other important information.

In the following, different popular and public music datasets are presented. They will be analysed according to their possible use in this research and how they meet the requirements specified above. After that, the reasons for the final choice of dataset are outlined.

4.1 Audio Set

In 2017, Gemmeke et al. published a dataset for audio events called Audio Set[1]. This is a large-scale dataset of audio events labelled by humans using a carefully structured hierarchical ontology of 632 audio classes. The source of the audio events is derived from 10-second segments of YouTube videos. These audio events are occurrences in the real world, like human sounds, animal sounds, natural sounds, sounds of things, source-ambiguous sounds, environment sounds and music. Then, each sound

[1] `https://research.google.com/audioset/`

is labelled with one ontology and a maximal hierarchical depth of six. In determining labels, the researchers ensured that the tags were intuitive and clearly distinct. [11]

For this project, it is a big advantage that all songs are publicly available in form of a YouTube link. For that reason, however, seammless integration is harder to accomplish because the individual songs have to be downloaded first. Another drawback is, that this dataset's focus is not on music but on general audio events. This is why, in the first step, the dataset would have to be filtered to contain only the music clips. However, the fact that the clips are only 10 seconds long means there is a risk that this is not enough stimulus to arouse the brain and see the distinct patterns. A further problem lies in the format of the data. Since the audio event labels range from many different occasions, only the audio events under the category of Music are important in this case. From this subset, only the songs with one of seven genre labels are required: pop, rock, hip hop, techno, rhythm and blues, vocal, and reggae. In total, there would be 40540 clips available. However, the songs are not balanced in regards of their genre. [3]

4.2 Million Song Dataset

The Million Song Dataset[2] is a collection of audio features and metadata for a million music tracks, which is also freely available. This is the largest research dataset in the field as it contains 1,000,000 songs, accumulating around 280GB of data. In addition to the songs, mainly from western commercial music in 2011, general metadata and audio features like pitch, timbre, etc., are provided using the EchoNest API. This API is an online resource supporting many music applications on the web. Since most of the "natively" available metadata comes from the EchoNest API, it is easily extendable to further labels like genres. [5]

Although this dataset may seem like a great fit because of its rich amount of songs, the number of songs is too much, since only 800-1000 songs are used during the acquisition, most of the dataset would have to be filtered out. Furthermore, not all songs are trimmed down to a shorter clip but are provided in full length, which does not meet the requirement for this study. Moreover, the genres would have to be added externally since they are unavailable in the native dataset. This makes the process of genre balancing much more difficult than with datasets where the labels are already present within the original dataset.

[2] `http://millionsongdataset.com/`

4.3 MagnaTagATune

One challenge when creating such datasets is determining a label or tag for each song. In the context of genres, how is it decided which genre a song belongs to, as there could be overlapping features. Based on this hurdle, Law et al. created a music dataset to evaluate different tagging algorithms called MagnaTagATune[3]. It contains 25,863 music clips, each 29 seconds long. These excerpts were taken from 5223 songs. These music pieces range across genres like Rock, Pop, Jazz, Blues, World, Metal, Punk and many more. Each clip is generally accompanied by a vector of binary annotations of 188 tags using the TagATune game. [14]

TagATune is a game with two players that aims to collect music tags from the players. Each player is presented with a music clip, which they are assigned to label. After that, both players see what the other player has chosen as tags. Based on this information, the players must decide whether they listened to the same music clip. [21]

From the perspective of this study, this music dataset has the advantage that each clip is around 30 seconds long, hitting the desired length for one stimulus. Furthermore, it is beneficial that the songs vary in the genres used, thus making them a good candidate for a music dataset. However, there are some drawbacks, especially lying in the labelling of each clip. Although the labelling process is well-thought-out and transparent, each song can have more than one label. Depending on the players, one clip can be annotated with more than one genre tag. And since there are in total 37 genre-related tags [26], deciding which genre is the most suitable choice for a song is very difficult. Related to that, since there are around 26,000 clips, one would have to remove a majority of songs to let it fit into the scope of this thesis. Combined with the fact that multiple genres can be chosen for each song, achieving a balanced dataset in terms of genre is difficult.

4.4 FMA Dataset

The Free Music Archive (FMA)[4] is an open dataset designed for multiple tasks in the domain of Music Information Retrieval (MIR). It consists of 106,574 tracks distributed over 16,341 artists and 14,854 albums. Although this dataset offers many different audio features and metadata about the songs, it is most tailored to music genres. Each track is arranged in a hierarchical taxonomy of 161 genres. As a labelling technique, the researchers labelled the song according to the opinion of the artist who sings the respective song. Therefore, subjective errors in tagging and labelling could be

[3] `https://paperswithcode.com/dataset/magnatagatune`
[4] `https://github.com/mdeff/fma`

minimised. In addition to that, all tracks are not only stored in simple .mp3 files, but also are fully licensed. To make the use of the FMA dataset easier, different subsets of the complete dataset are offered depending on the research question and scope. In total, there are four subsets one can choose from. The first one is called *Full*, which has every track untrimmed and is therefore unbalanced in genre. The *Large* subset contains the same tracks as the *Full* set, but now, the audio is limited to 30 seconds each, extracted from the middle of the track. To tackle the simpler problem of single-label prediction in genres, the *Medium* subset only contains those songs that have only one top genre, resulting in 25,000 30-second audio clips. Furthermore, the number of genres is reduced from 161 to only 16. However, this subset remains unbalanced. Defferrard et al. have constructed the balanced *Small* subset to solve this challenge. It holds 8000 tracks, each being 30 seconds long. From all the genres, the eight most popular ones were chosen, and based on this choice, 1000 clips were chosen for each genre. The eight genres in this case are Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop and Rock. [9]

The FMA dataset brings many favourable characteristics for this research. The *Small* Subset of this set is a perfect fit since it is balanced according to genres and does not contain that many songs. However, around 900 songs per genre would need to be removed to have a reasonable number of songs for the scope of this thesis. Another advantage is the recency of the publication. It is frequently used in different areas of MIR and is accompanied by many code examples and setup guides.

But, despite of being balanced in regards of the genres, the genres are not 100% clear resulting in unclear behaviour and characteristics of the songs. For instance, the genres International and Instrumental are not comprehensive, since many genres can either have a cultural impact or don't have any lyrics. Another disadvantage is that Classical music is not included, since this type of music is well-known for impacting the brain.

4.5 GTZAN Genre classification

Last but not least, there is the GTZAN Genre Dataset[5], which is a dataset specifically designed for MGR. There are 1000 songs in total originating from 2001 and 2002. Each track is trimmed to 30 seconds and ranges over 10 different genres. These genres are widespread, including blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae and rock. Furthermore, each genre contains 100 songs, resulting in a balanced song set in terms of genres. All audio was gathered from different sources like CDs, radio, microphone recordings and more to vary in the conditions of the recordings. Apart from the audio files, the dataset also contains audio features for each song. [30]

[5] `https://www.kaggle.com/datasets/andradaolteanu/gtzan-dataset-music-genre-classification`

Dataset	Number of songs	Average duration	Genres	Balanced option
Audio Set [11]	40,540	10s	7	no
Million Song Dataset [5]	1,000,000	219s	undefined	no
MagnaTagATune [21]	25,863	29s	undefined	no
FMA Small [9]	8,000	30s	8	yes
GTZAN [30]	1,000	30s	10	yes

Table 4.1: Overview of potential Music datasets. The attributes only correspond to the parts of the dataset that are interesting in terms of MGR. The term undefined is used when there are no clear genre tags or if they depend on subjective interpretation and individual preprocessing

As this dataset is specialised in studies regarding MGR, it is an excellent candidate for the thesis's purposes. With its 1000 songs, it fits perfectly in the right range of songs. Moreover, since each genre consists of 100 songs, it is perfectly balanced, meeting another requirement for the experiment. Another advantage is that the genres are clearly described, from which clear attributes can be derived. The only drawback in choosing genres is that some genres overlap in characteristics. For instance, jazz originates from blues, thus sharing a lot of common characteristics. Nonetheless, many of the other genres have apparent differences from one another, like classical and metal.

4.6 Final decision

All datasets presented in the previous sections have their strengths and drawbacks. An overview of the key attributes is shown in Table 4.1. Based on the evaluation of each dataset for the thesis's purposes, the most suitable song source for this work will be discussed and explained.

As mentioned, the audio set does not fit the requirements well. Since its focus is not on music but audio events, applying the set of clips in this use case is difficult. Moreover, ten seconds for each track is insufficient to arouse specific brain activity for each genre.

On the other end, the Million Song Dataset is not beneficial either. Although it only consists of music tracks, they are not trimmed and are full-length. Apart from that, 90% of all songs would have to be removed since only around 1000 tracks are necessary for this study, which can be a significant overhead (especially when ensuring a balanced set of songs).

From all the datasets, the MagnaTagATune dataset has the most sophisticated model

of tagging each clip, which tends to higher quality in labels. Another benefit is that all songs are trimmed to 29 seconds, meeting the duration requirement. However, this dataset is not focused on MGR and consists of 188 music-related tags. Moreover, the tags are binary vectors, so some songs could be multi-labeled, resulting in more difficult handling and unclear distinctions of songs and the genre they belong to. For this reason, the dataset was not chosen for this research.

Both the FMA and GTZAN dataset have very similar characteristics as they are specialised in MGR and therefore are balanced in genres, which is one of the most critical requirements in this study. Additionally, all clips are available as audio files with a constant duration of 30 seconds. The key difference lies in the size and genres present in the dataset. While the FMA set contains eight genres with 1000 songs each, the GTZAN consists of ten genres with 100 songs each. Besides, compared to the genres represented in the GTZAN dataset, the genres in the FMA collection are not very descriptive and (for some of them) difficult to distinguish. Therefore, the GTZAN song collection is chosen for the experiment setup and as a source of songs. Its easy file structure and naming convention can be seamlessly integrated into any setup, thus being the most suitable fit for this thesis.

5 Methodology

5.1 Acquisition

5.1.1 General setup, environment and stimuli

The experiment has been conducted in a laboratory setting. As mentioned in Chapter 4, the GTZAN Music dataset is chosen as a basis for the music stimuli. However, for the sake of the quality of the data, it would not be beneficial to hear all 1000 songs in one session. To avoid quick attention loss during a session and gather as much different data under different circumstances, the whole set of audio files is partitioned into 20 distinct sessions, each lasting around 30 minutes. This means that per session, 50 songs are played[1]. For each of these songs, not only are EEG signals measured, but also other biosignals like Photoplethysmogram (PPG), Inertial Measurement Unit (IMU) and temperature. However, this work will solely focus on the recorded EEG data. During each session, five songs from each genre are being played to ensure that each of the ten genres from the dataset is represented. This also ensures no genre bias in the data, thus resulting in a balanced signal across genres. During listening, the order of the songs is randomised to evoke a brain reaction that is as natural as possible, as the brain cannot predict the genre that is being listened to. Additionally, a five-second break is implemented in between songs to not overstimulate the brain.

All 20 sessions have been recorded over the course of two weeks, with an average of two sessions per day. Between each session, there was a short break of between ten and 30 minutes. This verifies a sufficient pause of the brain to fully prepare for the next session. The test person is the author, a 21-year-old, right-handed college student without any common (brain) diseases. To further guarantee a high quality in the acquired signal data, the experiment's environment and setting must be adapted. First and foremost, the test person is instructed to sit comfortably in a chair with his eyes closed to minimise artificats coming from blinking and eye-movement. Furthermore and most importantly, any movements, especially in the head region, must be limited and as minimal as possible since the EEG device reacts very sensitive to head movements. Apart from that, the room was quiet and isolated with dimmed natural lighting

[1] Because of a corrupt audio file, session ten only consists of 49 songs

combined with the brightness from a desktop monitor. The songs were played over the EEG headset (cf. Subsection 5.1.2) with average and constant volume.

5.1.2 Acquisition device

A non-traditional approach is chosen to record the EEG signal data coming from the brain. For this study, a prototype for a hybrid biosensing platform is chosen. It is a Bluetooth headset that can play music and measure biosignals at the same time. For connecting with a standard PC, it uses classic bluetooth and low-energy bluetooth for audio and biosignals respectively.

The headphones have multiple sensors and electrodes to measure biosignals like EEG, PPG, IMU and temperature. For EEG, the headphones are supplied with 2 × 4 sensor electrodes and two bias electrodes. The sensors are easily configurable and provide an ultra-low-noise front-end, meaning that the data points are amplified with minimal distortion and interference. Besides that, the electrodes sample with 24-bit and a frequency of 250 Hz.

For PPG, one optical sensor on the right side provides a high resolution of 19-bit ADC with a configurable sample rate. They also support raw data output or built-in algorithms to conclude heart rate or SpO2.

The IMU data is sensed by a 3-axis digital accelerometer fused with a 3-axis digital gyroscope. On a full scale, the accelerometer can recognize acceleration from $\pm 2\,g$ up to $\pm 16\,g$ (in steps of the power of two). The gyroscope can measure angular velocity from $\pm 125\,^\circ/s$ up to $\pm 4,000\,^\circ/s$. The recognised data is then combined to provide accurate information on orientation, movements and position.

Last but not least, there are three digital thermometers, one on the skin, another one inside the earcup and one on the outside of the earcup to measure ambient temperature. All of those temperature sensors have an accuracy of $\pm 0.1\,^\circ$C.

Since this headset is still a prototype, there still is ongoing development and research on the headset. Nonetheless, some small-scale research projects already employed the biosignals from the headphones. These projects range from stress detection using the PPG data to device control and the recognition of cognitive fatigue using the EEG sensors.

5.1.3 Technical setup

Multiple technical components have to come together to gather the experiment's data. All the components employ the Lab Streaming Layer (LSL). It provides a standardised way to stream time-sensitive data between devices and applications. Devices or apps

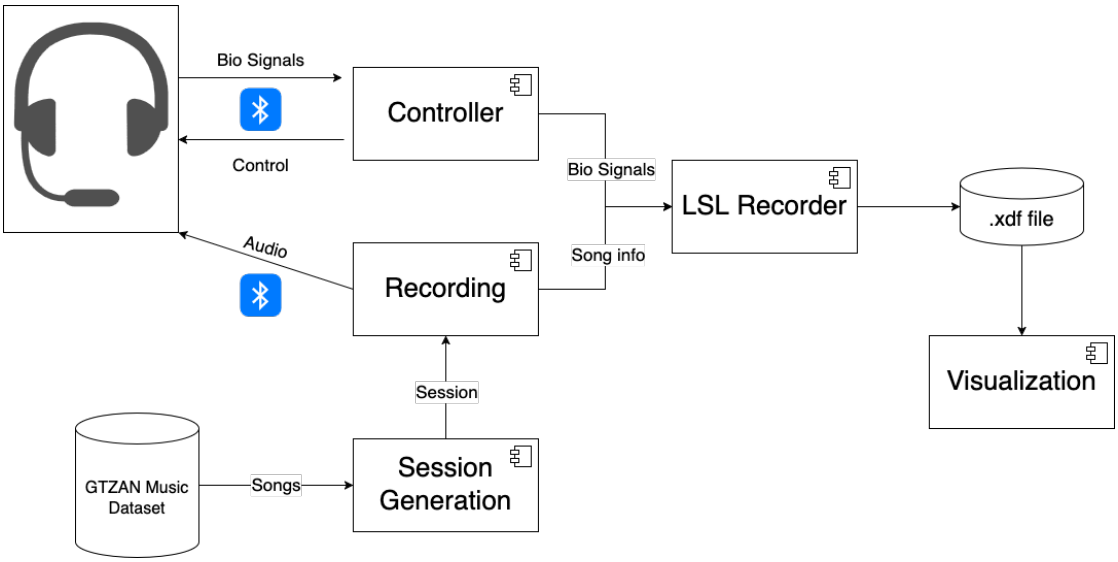


Figure 5.1: Technical setup

then push data on this stream, which is captured and stored in a special file form called .xdf. The workflow is illustrated in Figure 5.1 and will be explained in the following.

At the core, there are two main programs responsible for the data. On the one hand, there is the headset platform called acting as a Controller. This program is responsible for the sensor setup and starting and stopping the signal acquisition. Communicating with the headset's low-energy Bluetooth channel reliably sets up the biosensors of the hybrid headphones, making the device ready for acquisition. After a successful setup, the data is pushed on the corresponding LSL Stream. On the other hand, the study uses a custom Python script having two tasks regarding playing the music. Firstly, it loads all the songs from one of the 20 pre-generated sessions and randomly shuffles them. After that, the program plays each song one by one. Simultaneously, the script also pushes timestamps during recording to assign each segment to each song later.

By collecting all open streams on which the device/program push their data, the LSL recoder combines both components mentioned above. The collected information for each session is then stored in a seperated .xdf file, allowing for easy reading and visualisation.

5.2 Denoising and Preprocessing

After measuring and recording the EEG activity, the next step is to preprocess the acquired signal. Apart from segmenting the data into their respective song listened

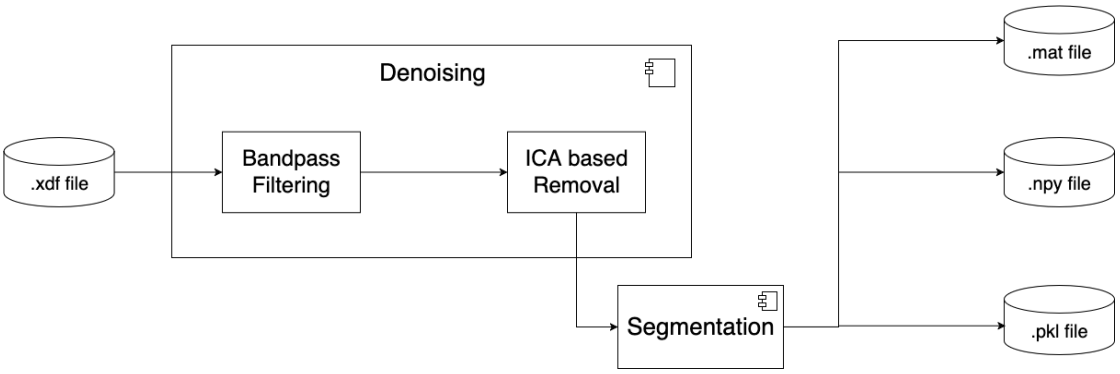


Figure 5.2: Preprocessing of the raw EEG signal

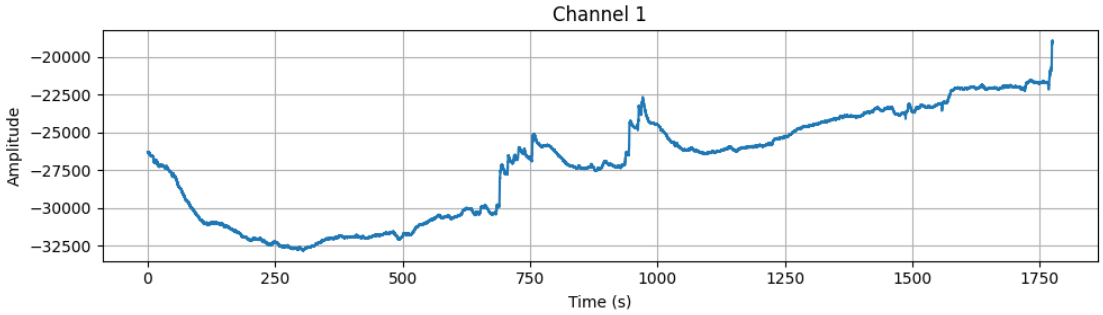


Figure 5.3: Raw signal of Session 1

to, the signal has to be denoised, which is the most essential step. As mentioned in Section 2.4, many different artifacts besides the neural activity appear during recording. Therefore, it is crucial to recognize and omit most of these contaminations to have valuable data for feature extraction and classification. The complete preprocessing pipeline is visualized in Figure 5.2, where each step is further explained in the following.

Before explaining how the data has been denoised, it is vital to investigate the state of the raw measured data. An example can be found in Figure 5.3. Most notably, there are significant amplitude shifts over the 30-minute recording sessions. Depending on the channel, there is either an increasing or a decreasing trend in amplitude. This can be reasoned with the fact that the electrodes are DC-coupled in the system. The drift then happens due to temperature changes or skin impedance.

To overcome this signal drift, the first step of the denoising pipeline is to apply a bandpass filter on the whole session. As boundaries, 1 Hz and 30 Hz are chosen as the lower and upper borders, respectively. This handles the mentioned drift and filters out all high and low frequency noise that does not match a certain brain rhythm.

Rejection Criterion	Threshold
Peak-to-Peak Amplitude (PTP)	90th percentile of all components
Kurtosis	90th percentile of all components
Skewness	90th percentile of all components
Autocorrelation (lag 1)	90th percentile of all components
Relative Delta Band Power	> 0.5 of total power
Relative Beta Band Power	> 0.35 of total power
Mixing Matrix Sparsity	< 3 active channels (relative threshold = 0.2)

Table 5.1: ICA Component Rejection Criteria

After removing most of the frequency-related noise, the artifacts in the EEG data must be detected and removed. An ICA algorithm is applied to the session data. After prewhitening the data, an ICA is applied, splitting the EEG matrix into a mixing matrix accompanied by the sources. Since eight EEG channels are present, the ICA algorithm was performed with eight components. As a next step, it has to be decided whether a component is an artifact or an actual source of cerebral activity. This thesis deals with an unconventional way of measuring the EEG data, so custom criteria must be determined for this task. In Table 5.1, the rejection criteria for each component can be seen. PTP, skewness, kurtosis, and autocorrelation are used to detect unusual signal characteristics deviating from brain activity. The spectral power thresholds, including the Delta and Beta Power ratio, find artifacts by finding components dominated by specific frequency bands. Lastly, by investigating the sparsity of the mixing matrix, it is evaluated how widely a component affects the EEG channels. A component is classified as an artifact if at least two of the mentioned criteria are met. The components labelled as artifacts are then set to zero before converting the mixing matrix with the now cleaned sources to the artifact-removed version of the original EEG data. This results in a signal where most of the high amplitudes are removed with a much more consistent EEG signal, thus suggesting that most of the unwanted artifacts have been removed.

With the cleaned EEG signal at hand, one can now segment the signal into its 50 individual songs[2]. Since the start and the end of one song are pushed to the LSL stream, the timestamp of those markers is read out to only keep the signal data appearing in this interval. This is done over each channel, thus resulting in a table with each song occupying a row and the channels as columns.

After all these steps, the denoised and segmented data is stored in three different

[2]The custom Python script also supports the segmentation of biosignals other than EEG. However, these will be ignored in the following as this thesis focuses on EEG signals

formats, being .mat, .npy, and .pkl, where each session is held in one file across all file formats.

5.3 Feature extraction

The next step after the signal preprocessing is the extraction of meaningful features used for the classification task of predicting genres. Finding the right features is a difficult task in this particular use case. This is mainly because the field of MGR based on EEG is pretty young, lacking in recent studies. Another problem arises from the fact that an unconventional EEG acquisition device is used with non-standardized electrode placement, on which little research exists.

To overcome those obstacles, it is essential to compute as many different features across all the domains as possible to have a versatile range of different characteristics. An optimal feature subset can be chosen using feature selection techniques, allowing for sufficient classification results.

For all features extracted from the channels, the first four seconds are taken into account. This way, the activation of the stimulus is captured.

As mentioned, features are computed across all three domains (the time domain, the frequency domain, and the time-frequency domain). All features are computed per EEG channel, thus leading to a feature space of $n_{channel} \times n_{features}$ different parameters.

An overview of all features extracted in the time domain is given in Table 5.2. The features calculated are a mix of classic statistical measures like mean, maximum, and minimum, and tailored signal metrics like entropy, RMS, and the Hjorth parameters.

To gain knowledge about the signal's time properties, features from the frequency domain have also been extracted. For all features described in Table 5.3, Welch's method was applied as a basis to determine the PSD. To balance both computational cost and accuracy, a 2 s window is used. With the PSD at hand, the spectral entropy is calculated to measure its complexity and irregularity, which can provide information about different mental states. More importantly, the relative power of each brain rhythm frequency band is calculated to further depict the neurological condition of the brain during each session. The brain rhythm's frequency bands are defined as follows:

- Delta: $0.5 - 4$ Hz
- Theta: $4 - 8$ Hz
- Alpha: $8 - 13$ Hz
- Beta: $13 - 30$ Hz

Feature	Description
max	Maximum value of the signal
min	Minimum value of the signal
std	Standard deviation of the signal
mean	Mean value of the signal
median	Median value of the signal
var	Variance of the signal
range	Peak-to-peak range of the signal
skew	Skewness of the signal distribution
kurtosis	Kurtosis of the signal distribution
rms	Root mean square value of the signal
zcr	Zero crossing rate
h_mobility	Hjorth mobility parameter
h_complexity	Hjorth complexity parameter
energy	Total energy (sum of squares) of the signal
shannon_entropy	Shannon entropy of the signal

Table 5.2: Extracted Time-Domain Features

The relative power is taken to normalize the data across all sessions to account for the session-dependent measurements. In addition to the relative band power, ratios between specific rhythms can not only further normalize the results but also give a relative measure of brain state changes. This also highlights proportional changes in frequency bands, being possibly more informative than the absolute power.

To combine the time and frequency domains, a DWT calculates the features mentioned in Table 5.4. In that case, the Daubechies wavelet is used as the mother wavelet function, and a level of four is chosen, resulting in five subbands. Time domain features are extracted from each subband, including statistical measures and signal-specific parameters.

As already hinted, total features depend on the number of EEG channels, thus leading to high dimensionality. An MRMR feature selection algorithm using MID is used to choose the optimal feature set out of all calculated parameters. For both Relevance and Redundancy, the mutual information value is utilized to calculate both measures. The number of selected features is variable and can be optimized depending on the classification task or the ML model applied.

In conclusion, a rich set of versatile features provides information about many facets of the EEG signal. Furthermore, an MRMR feature selector is applied to improve the

Feature	Description
spectral_entropy	Shannon entropy of the normalized power spectral density
DELTA_relative	Relative power in the delta band
THETA_relative	Relative power in the theta band
ALPHA_relative	Relative power in the alpha band
BETA_relative	Relative power in the beta band
GAMMA_relative	Relative power in the gamma band
ALPHA_BETA_ratio	Ratio of alpha power to beta power
THETA_ALPHA_ratio	Ratio of theta power to alpha power
THETA_BETA_ratio	Ratio of theta power to beta power

Table 5.3: Extracted Frequency-Domain Features with Welch's method

Feature	Description
tfd_band_i_mean	Mean of wavelet coefficients in subband i
tfd_band_i_median	Median of wavelet coefficients in subband i
tfd_band_i_std	Standard deviation of wavelet coefficients in subband i
tfd_band_i_variance	Variance of wavelet coefficients in subband i
tfd_band_i_entropy	Shannon entropy of normalized wavelet energy in subband i
tfd_band_i_relative_energy	Relative energy of subband i with respect to total wavelet energy
tfd_band_i_skewness	Skewness of wavelet coefficients in subband i
tfd_band_i_kurtosis	Kurtosis of wavelet coefficients in subband i

Table 5.4: Extracted Time–Frequency Domain Features with a level 4 DWT, where i is the i-th subband

quality of the feature set during classification.

5.4 Classification

Many difficulties appear when it comes to classification, as it is the most challenging step in the EEG pipeline. It is particularly difficult within this thesis out of two reasons. Firstly, research on EEG based MGR is minimal, so state-of-the-art methods for this area do not exist. Additionally, one of the only reference papers achieved mediocre performance on single EEG classification on the DEAP dataset on subject-level [12]. Secondly, the fact that the data was acquired using an unconventional method for which no related work exists makes it hard to verify whether the gathered data is even suitable for such a classification task.

To overcome those hurdles, MGR is performed on the thesis's handcrafted dataset and the well-known DEAP dataset. Since this is a common source for EEG based classification and already provides genre labels provided by [12], it can be used as a source of truth in this specific classification task. Therefore, the Classification methodology can be split into two parts. The first part involves performing MGR on the DEAP dataset to research and find out what works best and what doesn't yield desirable results. This knowledge and usable results from the first part are then transferred to the Headset data. In this way, whether the same methodology that works for a state-of-the-art dataset also applies to the Headset data can be investigated leading to faster results, and a hint at the signal's quality.

5.4.1 DEAP

As already said, the publicly available preprocessed DEAP dataset is used as a signal source for classification. However, the songs are not labeled according to genre, the genre tags applied by [12] are taken into account. In this way, it is easily possible to compare the results to current research in EEG based genre recognition. This results in an imbalanced dataset because of the 40 songs, 60% belong to the POP genre and 40% to the ROCK label.

From the available 32 EEG channels, multiple features have been precomputed from each channel to depict the neural activitiy accordingly. The features are chosen based on past research using this dataset[3]:

- Alpha, Beta, Theta, Delta, Gamma relative band power
- Alpha-Beta ratio
- Theta-Alpha ratio
- Theta-Beta ratio

[3] cf. Section 3.4

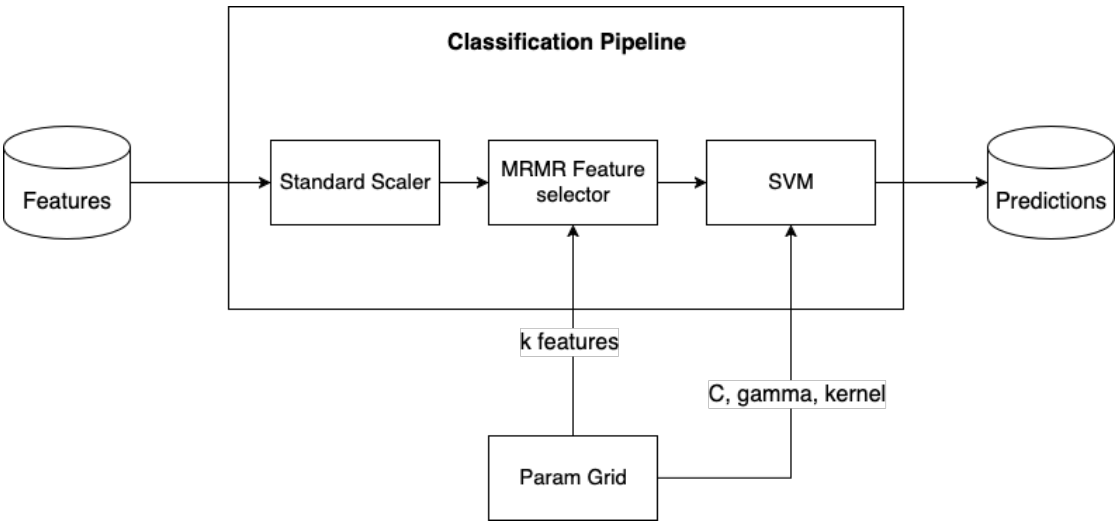


Figure 5.4: Classification Pipeline for Hyperparameter Tuning and Evaluation

- Entropy from each subband from a level 4 DWT

Important to note is that the features were calculated for the first four seconds of the recorded signal to capture the activation impulse of the stimulus. The methods used for extracting the signal are the same as mentioned in Section 2.5

Regarding classification, a sophisticated, modular ML pipeline is employed to get sufficient results, allowing for easy embedding in Hyperparameter tuning techniques like Grid search or Randomized search, which is illustrated in Figure 5.4. At first, a z-normalization is applied to each feature, so that the data have zero mean and unit variance. This step is important since it accounts for abnormalities and unifies the data, leading to better classification results. An MRMR feature selection method is utilized to choose the optimal subset of features for this task. This also reduces the dimensionality, which is important since the number of samples is small (40 samples per subject).

The selected subset is then fed into an SVM with balanced class weight and a constant random state to make the model's outputs deterministic over multiple iterations.

In this case, a subject-based classification approach is followed. In this thesis, an individual model is optimized for each subject allowing for a more subject-centric classification approach.

A Grid search is embedded as well to find the optimal configuration to get the best performing model possible. Apart from the SVM's hyperparameters (C, γ, kernel), the number of features selected by the MRMR is also tuned. An overview of the hyperparameter space is illustrated in Table 5.5. The distribution of the optimal parameters can then be further analysed to find out which values perform best in the

task of MGR for this dataset.

To evaluate the model's performance, the primary metric used are the mean accuracy with the balanced accuracy between all subjects. Alongside that, the f1-score is taken into account to have a measure for both precision and and recall in an imbalanced class environment. To get an idea of prediction distribution, the confusion matrices of each optimal model are also computed per subject. Moreover, the mean of those metrics over all subjects is calculated as well to gain information about overall performance and similarities. In the case of the confusion matrices, a global confusion matrix is generated by unifying all predictions and test labels. Those metrics allow for a detailed analysis of MGR in the case of the DEAP dataset.

Parameter	**Search Space**
selector__k	{2, 5, 8, 10, 15, 20, 30, 50}
svm__C	{0.001, 0.01, 0.1, 1, 10, 100, 1000}
svm__kernel	{linear, rbf, sigmoid}
svm__gamma	{scale, auto, 0.0001, 0.001, 0.01, 0.1, 1, 10}

Table 5.5: Hyperparameter search space for SVM with MRMR feature selection.

5.4.2 Headset data

To compare the results of the DEAP dataset with the data originating from the hybrid EEG headset, the same methodology described in Subsection 5.4.1 is employed, incldu-ing the Classification pipeline and the hyperparameter space in Figure 5.4 and Table 5.5 respectiveley. Nonetheless, there are some important differences that will be mentioned in the following.

Firstly, the number of channels is reduced compared to the DEAP dataset which was measured with a professional EEG device. For this reason, and because no related work exists for the EEG headset, all features mentioned in Section 5.3 are available in the feature space from which the MRMR feature selector can choose from.

Secondly, in contrast to the DEAP dataset, the Headset dataset is balanced in terms of genres as each session contains five songs from each possible genre (ten in total). Therefore, the balanced accuracy is not taken into account as this metric mostly makes sense for imbalanced datasets. However, the f1-score and confusion matrices are still analysed to have a metric for how well the model performs on multi-class prediction.

6 Results

6.1 DEAP

6.2 Headset Data

7 Conclusion

7.1 Key findings and results

7.2 Problems and Limitations

7.3 Future work

Abbreviations

fMRI functional Magnetic resonsance imaging

ML Machine Learning

BMI Brain Machine Interface

CNS Central Nervous system

AP Action potential

ADC Analogue to Digital converter

BCI Brain-Computer Interfacing

BSS Blind Source Seperation

MARA Multiple Artifact Rejection Algorithm

EMG Electromyography

ECG Electrocardiography

EOG Electrooculography

PCA Principal Component Analysis

ICA Independent Component Analysis

HOS Higher-order statistics

DL Deep Learning

CNN Convolutional Neural Network

XGB Extreme Gradient Boosting

SVM Support Vector Machine

MFCC Mel Frequency Cepstral Coefficients

DAG Directed acyclic graph

PTP Peak-to-Peak Amplitude

FMA Free Music Archive

MIR Music Information Retrieval

EEG Electroencephalography

PPG Photoplethysmogram

IMU Inertial Measurement Unit

LSL Lab Streaming Layer

MGR Music Genre Recognition

MER Music Emotion Recognition

DEAP Database for Emotion Analysis using Physiological Signals

MEG Magentoencephalography

STFT Short time Fourier Transform

RMS Root Mean Square

ZCR Zero Crossing Rate

PSD Power Spectral Density

DFT Discrete Fourier Transform

TFA Time-frequency Analysis

CWT Continous Wavelet Transform

STFT Short Time Fourier Transform

DWT Discrete Wavelet Transform

MRMR Minimum Redundancy Maximum Relevance

SEED SJTU Emotion EEG Dataset

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